

Adam Harder

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The latest version of this document is available at: <https://github.com/adamharder/resume/blob/master/resume.pdf>

SUMMARY

Senior software engineer and technical lead with 20+ years across embedded systems, cellular/mobile security, and vulnerability research. Leads 5–6 person research teams end-to-end: technical direction, sprint planning, direct technical contributions, and customer-facing delivery. Deep hands-on background in full-stack development (Python, Postgres, JavaScript/TypeScript), reverse engineering (ARM, C/C++, CAN bus).

EMPLOYMENT HISTORY

Nightwing

2019 - Present **Technical Lead** — Led 5–6 person research teams across multi-year programs. Owned technical direction, sprint planning, and direct technical contributions; delivered regular status to the customer; provided performance feedback for all team members and owned team composition.

- **2019 – 2021 — Firmware CVE tracking.** Lead developer for a web crawler that collected firmware images across ~12 vendors, building a ~2 TB corpus used to track CVEs across generations of firmware. (Python, Postgres)
- **2021 – 2024 — Industrial robotics vulnerability research.** Led a team reverse-engineering compiled C/C++ + industrial robot systems, including live instrumentation of running systems via the CAN bus.
- **2024 – Present — Physical access-control vulnerability research.** Leading reverse engineering and vulnerability research of building security systems — HID readers, NVRs, and IP cameras. (Python)

Independent Contractor

2016 - 2019 * **Raytheon Intelligence Information Services** * Developed a software build system and mirroring infrastructure for an air-gapped network.

- **Facilities Survey** — Lead developer for the CMS View and Space Manager hospital facilities management product suite (JavaScript, TypeScript, Python backend on Microsoft Azure). Multiyear effort spanning multiple disciplines.
- <https://www.fsiservices.com/cms-view>
- <https://www.fsiservices.com/cms-space-manager>
- **LandEconics** — Developed a cost projection estimation system for landscape architects.

Endgame

2011 - 2016 **Director, Mobile Access**

- Led a three-engineer team developing repeatable, isolated, over-the-air black-box test procedures for an unsupported third-party smartphone app, identifying and characterizing bugs across multiple software versions.
- Led a three-person, six-month sub-API-level study of smartphone sandbox limitations across iOS, Android 4.2, Symbian, WP8, and BlackBerry 7.
- Internal research projects repurposing consumer electronics hardware; developed an Android wireless sniffing tool including the front-end implementation.

Stack: ARM architecture, Java, Python, RealView ARM Tools, IDA Pro. Platforms: Nokia/Symbian, Android, Windows Phone 8, BlackBerry 7. Test equipment: Anritsu, ip.access Picocell, OpenBTS. Waveforms: 802.11, GSM, WCDMA/UMTS, CDMA.

Raytheon S.I. Government Solutions

2008 - 2011

- Various embedded development projects focused on mobile phone platforms.

- Implemented a complete ARMv6 disassembler using test-driven development with a representative instruction unit test bed of 250,000 test cases.
- Worked with a small embedded team to repurpose a hardware debugging protocol into a data analysis tool, make calls into non-standard system drivers, and perform black-box testing and security auditing of third-party software.
- Led a team of 8 engineers to develop a cell-phone-agnostic security test framework. Developed customer relationships that leveraged these successes to open new business areas.

Stack: ARM architecture, C/C++, Python, RealView ARM Tools, IDA Pro. Platforms: Apple, Nokia/Symbian, Google. Test equipment: OsmocommBB, Anritsu, Rohde & Schwarz, Agilent, OpenBTS. Waveforms: 802.11, GSM, WCDMA/UMTS, CDMA.

Harris Corporation

November 2006 - September 2008

- Sole developer retrofitting the software architecture of a GSM product for the UMTS protocol, using the OSS Nokalva ASN compiler to generate libraries for parsing UMTS Layer 3 messages. Provided on-site customer test support and product demonstrations domestically and abroad.
- Led a team of 2–3 developers to productize the UMTS prototype and support customer prototype testing; additionally built a parser/compiler that generated C code from ASN specification documents. First exposure to project scheduling, cost estimation, and team leadership.
- Stack: C++, MFC, Windows API, Visual Studio 6 & 2005, SQLite, Apache ANT, Xoreax IncrediBuild, InnoSetup, GOLD Parser, OSS Nokalva ASN C++ tools; ASN, CSN, UMTS, GSM, Doxygen.

April – October 2006

- Carried a design through Preliminary and Critical Design Review for a set of enhancements to a GSM product.
- Implemented the UI for a modem-based SMS product (interfaces to the engine changed several times over the course of development). Stack: C++, MFC, Windows API, Visual Studio 6.0 & 2003, ChartFX/Microsoft Chart COM components, U3-compliant USB package; Multi-tech and Sierra GSM modems (PCMCIA & USB).

EDUCATION

Auburn University — Master of Software Engineering, August 2005

Baylor University — Bachelor of Science (Computer Science), December 2002

TALKS & PUBLICATIONS

Malware Technical Exchange Meeting 2015

Exploitation of Mobile OEM Software

DOJ National Security Division's Cyber Awareness Series 2015

Mobile Security Backgrounder

Towards an Indoor Location System Using RF Signal Strength in IEEE 802.11 Networks

Adam Harder, Lanlan Song, Yu Wang International Conference on Information Technology Coding and Computing, ITCC 2005, Sponsored by IEEE Computer Society, April 4-6, 2005, Las Vegas, NV